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Modified Invasive Weed Optimization for MPPT of PV Powered Water Pumping System Driven by BLDC Motor



Abstract: - Water Pumping Systems (WPSs) are must be required in human civilization, serving various purposes including agriculture, drinking, and industrial applications. In rural areas, reducing the strain on the grid and providing electricity to WPS can be achieved through the implementation of a locally established standalone Photovoltaic (PV) fed WPS unit. Brushless DC motor (BLDCM) is highly suitable for WPS due to their numerous advantages. However, the addition of batteries to the system can lead to increased costs and maintenance requirements, making it unaffordable for farmers. Therefore, a WPS driven by BLDCM fed by PV system without batteries is considered in this paper, and a sensor-less speed control unit is also incorporated. The power generated by PV unit can be affected by partial shading conditions (PSCs), resulting in suboptimal power production when using traditional Perturbed and Observe (P&O) algorithms. To address this issue, the Modified Invasive Weed Optimization Algorithm (MIWOA) is hybridized with the P&O method. The Hybrid MIWOA – P&O method proves to be effective in maximizing power extraction from PV systems under both PSCs and normal conditions. Comparative analysis with GA, PSO, and GWO is also conducted. To minimize system costs, the single stage operating WPS is implemented in this article. This eliminates the need for an additional DC-DC circuit for the operation of MPPT since inverter is used as a MPPT purpose. Comprehensive findings have been showcased through the utilization of OPAL-RT to verify the suggested system in both stable and changing conditions.

Keywords: PV, Water Pumping System, BLDC, MIWO Algorithm, Sliding Mode Control.

Nomenclature:

Photovoltaic: PV.

Maximum Power Point Tracking: MPPT.

Perturbed and Observe: P&O.

Partial Shading Effect: PSE.

Water Pumping System: WPS.

Genetic Algorithm: GA.

Particle Swarm Optimization: PSO.

Grey Wolf Optimization: GWO.

Hardware – in the – Loop: HIL.

Sliding Mode Control System: SMCS.

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Takagi-Sugeno Fuzzy Controller: (TSFC)
 Permanent Magnet Synchronous Motor: PMSM.
 Brushless DC motor: BLDCM.
 Model Reference Adaptive Controller: MRAC.
 Sensor-less Speed Controller: SSP.
 Induction Motor: IM.
 DC-link control: DCLC.
 Local Maximum: LM.
 Global Maximum Power Point: GMPP.
 Hardware-In-Loop: HIL.

I. INTRODUCTION

Water is a vital necessity for plants and animals alike. The utilization of pumps in delivering water serves a critical function in various sectors such as drinking, agriculture, and industrial processes. The availability of water greatly impacts agricultural yield and this issue become more pronounced during the summer season due to water scarcity. Water supply is not only necessary for agriculture, but also for human consumption and industrial needs. Typically, WPSs consume a significant amount of energy or electrical power to meet these demands. However, during the summer season, there is ample opportunity to harness solar energy and convert it into electrical power through PV cells. This has led to the increasing use of solar water pumps, especially in rural and remote areas where power supply is unreliable. Solar powered pumps not only help reduce carbon dioxide emissions, but they are also noise-free, making them environmentally friendly. Additionally, PV powered WPSs are viable solutions for supplying water in apartments where there is sufficient rooftop space for PV system installation. Several research projects have been carried out to enhance the design, functionality, effectiveness, and affordability of solar-powered WPSs. In contrast to motors, diesel engines are expensive and have negative environmental impacts. Moreover, the feasibility of extending electricity access to remote regions is frequently limited in numerous global locations. There are around 21.0 million irrigation pumps, with over 9.0 million being fueled by diesel engines, while the rest are operated using the primary power grid [1-5] in India. In various locations worldwide, numerous houses and industries utilize WPSs that are powered by either diesel or the power grid. As a result, the grid experiences an increase in load demand when motors are used in WPSs. In such situations, it becomes cost-effective and environmentally friendly to install a locally positioned WPS that is fed by a PV system.

PV-based WPSs have gained popularity in numerous countries recently. A PV fed WPS without battery bank offer a maintenance-free and cost-effective solution. However, to maximize energy harvesting from the PV unit and ensure optimal utilization by PV panels, a MPPT circuit with a suitable controller is necessary. Nevertheless, incorporating an additional device for MPPT will raise the overall size as well as cost of the WPS. Therefore, single-stage PV powered WPSs can present a compact and economical alternative. In addition to DC, AC motors are highly suitable for driving pumps to lift water from significant depths. Researchers have suggested different kinds of AC motors for photovoltaic-based water pumping systems [6-10]. For instance, in some studies, authors have suggested WPSs where the pump is powered by an IM. In another study, a WPS with a synchronous reluctance motor was proposed, along with the implementation of a various investigation control methods to work under PSCs for MPPT. Furthermore, a PI-fuzzy controller was proposed for a WPS coupled with a PMSM. Notably, authors hasn't considered the impact of PSCs. Authors have also introduced a grid connected WPS driven by IM and powered by PV, where surplus energy delivered by the PV unit is fed back to the grid when pumping water is unnecessary. Moreover, sensor-less control method based on ANN has been utilized for IM control method by field-oriented to operate the pump for water extraction from underground sources. In [11], the authors successfully developed a grid integrated WPS driven by a BLDCM. Furthermore, the identical inverter was employed to improve the power integrity of the electrical network. The BLDCM was chosen due to its numerous advantages over other types of motors, making it highly suitable for WPS applications, particularly in single stage model [4, 11-12]. Therefore, this paper focuses on utilizing the BLDCM for the implementation of a WPS powered by PV energy.

There are several methodologies available to work as MPPT of PV unit. Among these algorithms, the P&O mechanism is gaining attention from scholars due to its high efficiency and simplicity [2-4, 13]. In a PV unit,

multiple cells are arranged in parallel and series combinations. As a result, PSC can occur in the PV system due to various factors such as clouds, dust, buildings, trees, birds, and clouds [13-14]. The conventional P&O method fails to work as MPPT algorithm under PSCs because of the presence of multiple LM power points [13]. Hence, it is essential to incorporate an optimization method into the P&O method in order to pinpoint the GMPP. Numerous researchers have applied MPPT strategies to PV based WPS. In [15], a synchronous reluctance motor driven WPS powered by PV under PSCs was developed by authors. In [16], the authors suggested a grid tied PV for PMSM coupled WPS and introduced sensor-less field-oriented control. The utilization of a boost converter for MPPT in the PV system was done by the authors in [16], without taking into account the influence of PSCs. On the other hand, in [17], a distributed MPPT technique was suggested by the authors to enhance conversion efficiency by employing multiple converters. Nevertheless, this approach leads to increased costs and system size, with no mention of addressing partial shading in [17]. The impact of PSC on WPS driven by IM in a PV system is analyzed by the authors in [18]. They found that high variations in the voltage at dc-link caused by PSCs can negatively affect the power converter's efficiency, motor lifespan, and water pump output.

In addition to various optimization techniques, the MIWOA is widely used and can offer an efficient solution for quickly identifying the GM point [19-20]. This study introduces the hybrid MIWOA – P&O algorithm to find the MPPT level under PSC. The method under consideration is evaluated against GA [21], PSO [22], and GWO [23] in order to enhance the system's efficiency within the defined PSCs. A well-designed control strategy of the inverter is essential for effectively driving the BLDCM. Monitoring the velocity of the motor-pump unit located beneath the water's surface presents difficulties and comes with a high price tag. Hence, this study proposes the utilization of a sensor-less speed controller implemented through the MRAC method. Furthermore, a SMCS is incorporated to regulate the speed of the motor based on the power availability of the PV unit. The BLDCM speed is regulated by adjusting the voltage at dc-link of the inverter to align with the desired value generated by the recommended MPPT method, as illustrated in Figure 1 in the DCLC section. Compared to a PI controller, the TSFC is more effective in rapid changes of irradiances for regulating the voltage at dc-link [24]. Therefore, a TSFC is implemented in this study to manage the dc-voltage.

II. SYSTEM OVERVIEW

Wherever Times is specified, Times Roman or Times New Roman may be used. If neither is available on your word processor, please use the font closest in appearance to Times. Avoid using bit-mapped fonts if possible. True-Type 1 or Open Type fonts are preferred. Please embed symbol fonts, as well, for math, etc. The depicted WPS shown in Figure 1 includes a PV unit for supplying electric energy, an inverter for obtaining AC from DC and serving as an MPPT converter for the PV unit, a BLDCM for operating the pump to raise underground water, and a cutting-edge control method for the inverter. Since the pump is submerged, sensing the speed of BLDCM is a challenging task. A speed sensorless control strategy was devised utilizing MRAC to tackle this problem, while a SMCS was integrated to monitor the speed matching the desired value produced by the DCLC. The following sections provide a detailed analysis of these components.

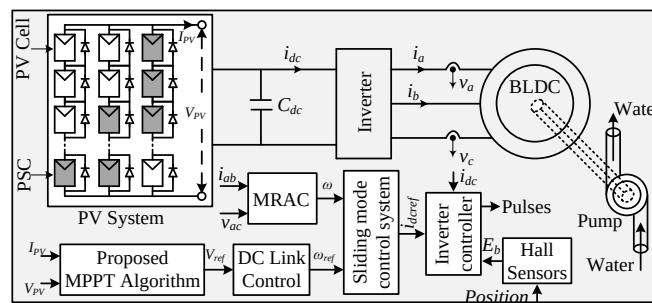


Figure 1: Proposed PV fed WPS.

A. WPS [25]

Normally, the breakaway torque required for a motor is around 15%–30% of the rated torque to surpass the static friction when lifting water. Once the speed reaches a certain threshold (ωt), the pump begins to deliver the water. The water flow rate (Q , gal/min) is directly proportional to the speed of the pump (ω), as indicated in equation (1)

obtained through curve fitting [25]. Furthermore, the system's leader is controlled by a non-linear equation denoted as equation (2).

$$Q = \begin{cases} a\omega - b & \omega \geq \omega_t \\ 0 & \omega < \omega_t \end{cases} \quad (1)$$

$$H = C_0\omega^2 + C_1\omega Q + c_2Q^2 \quad (2)$$

The motor's required horsepower capacity is determined through calculation.

$$W_{hp} = \frac{Q \times H}{3960} \quad (3)$$

Where, c , c_0 , c_1 , c_2 and b used as constants and horse power by W_{hp} , 'H' in feet denote total dynamic head.

The power needed to operate a pump must exceed the W_{hp} due to the pump's efficiency. The horse power needed at the pump a specific flow rate against a specific H is known as the brake horsepower (BHP), as stated in equation (4).

$$BHP = W_{hp} / (\text{efficiency of pump} \times \text{efficiency of electric drive}) \quad (4)$$

Assuming $Q_{\max}$ is 150gal/min, H is 50m, pump efficiency is 1, and efficiency of electric drive is 0.95, BHP requirement according to equation (4) is 4876 W. The solar panel's maximum power rating is chosen as 4.81kW, in accordance with the maximum BHP and the recommendation provided in [26] for PV fed WPS. Based on equation (2), the motor's maximum speed is determined to be 190rad/s, which is roughly equivalent to 1800rpm. The solar panel's power availability will be around 1.9kW if the minimum irradiance is 400W/m². The solar panel's minimum power will allow the motor to operate at a minimum speed of 720rpm. The load torque on the motor is estimated at 25 Nm.

B. PV System [2-4, 14, 21-27]

The choice of PV unit is determined by the power needed by the motor, as outlined in the preceding section. A total of 16 PV modules, each with a capacity of 302W, have been chosen to generate the necessary maximum power of 4800W. Please consult Table-1 for a thorough compilation of PV module specifications.

Table-1: a Module of PV Parameters [27]

S.No	Parameters	Values
1	Short circuit current (A).	8.16
2	Open circuit voltage (V).	48.50
3	V_{MPPT} .	39.60
4	I_{MPPT} .	7.66
5	Maximum power.	302.0
6	Number of series modules.	16

With 16 modules arranged in series to generate the necessary voltage at dc-link and required power, the likelihood of PSCs increases. The PV system has been extensively modeled under PSC conditions based on references [2-4, 13-14, 21-24]. The voltage at the MPPT can be accurately traced using the conventional P&O method, as shown in the mathematical expression below for consistent solar irradiance.

$$V_{mpp}(j) = V_{mpp}(j-1) + \Delta V \times \text{sign}\left(\frac{dP_{pv}}{dV_{pv}}\right) \quad (5)$$

Three distinct PSCs were examined in this research, as detailed in Table-2, accompanied by the power versus voltage curves showed in Figure 2. The MPPT levels with the corresponding voltages are also indicated in Figure 2. During PSCs, the PV unit can demonstrate multiple LM points and one GMPP. Traditional P&O algorithms are capable of tracking any of the LM points. However, to identify the GMPP, an efficient optimization technique

must be employed. Several researchers have explored optimization methods for PV unit operating under PSCs. For instance, in reference [21], GA was utilized for MPPT under PSCs of PV systems. In reference [22], the PSO technique was implemented for MPPT. Additionally, in reference [23], the GWO method was proposed for PV unit under PSCs. Nevertheless, the MIWOA has garnered significant attention from researchers due to its rapid convergence compared to other methods such as GWO, PSO and GA, [19-20]. The combination of P&O and MIWOA can offer a best solution for achieving the MPPT under PSCs in photovoltaic systems.

Table-2: PSCs list

Pattern	Irradiance levels
PSC-1	Modules (W/m ²) 1-2: 1000, Modules: 3-7: 904, Modules: 8-12: 820, Modules: 13-16: 701.
PSC-2	Modules (W/m ²) 1-4: 995, Modules: 5-8: 930, Modules: 9-13: 724, Modules: 14-16: 550.
PSC-3	Module (W/m ²) 1: 822, Modules: 2-5: 710, Modules: 6-12: 552, Modules: 13-16: 301.

PV modules are comprised of numerous PV cells, and the PV system/array is setup by arranging 16 modules in series. Consequently, the likelihood of encountering PSC is significantly elevated. Regardless of the unpredictable weather situations, the proposed method for the device should be capable of handling the PSC impact and efficiently capturing the maximum power available. In the P&O algorithm, the perturbation direction is determined by sign of power change.

In the P-V curves in the case of PSC, multiple local peaks are typically observed, but there exists a global MPP (GMPP) in the curve, as depicted in Figure 2. Traditional MPPT methods usually target the nearest local maximum point, as they are unable to differentiate between the local and global maxima. Therefore, the MIWOA method combined with the P&O mechanism helps improve GMPP search ability and achieve random convergence with minimal fluctuations during PSCs. Additionally, the positions of GMPP and LM points are randomly influenced by changes in PSCs and the configurations of PV unit, as illustrated in Figure 2.

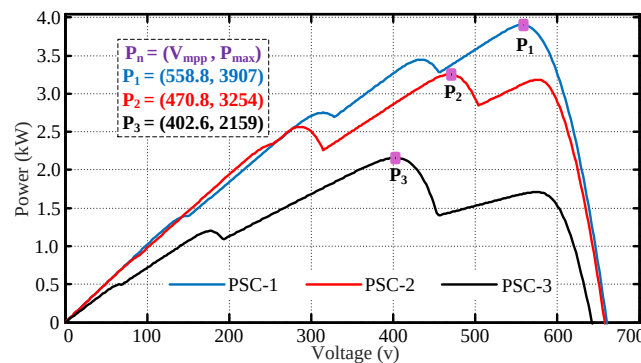


Figure 2: PSCs listed in Table-2 P-V characteristics

The combination of the P&O and MIWOA can showcase their strengths through their innate search capabilities. The MIWOA, once implemented, ensures the convergence necessary to effectively achieve the GMPP due to its long-term response. Conversely, the P&O algorithm cannot promise convergence to attain the GMPP. Nonetheless, it is acknowledged for its swift reaction in uniform irradiance settings. The concept involves relocating all the plant's weeds to bring the system closer to the GMPP, allowing the P&O algorithm to then maintain proximity to the GMPP. In the absence of PSC, the MPPT method is executed by the P&O method. If the PSC is detected by the system, the MIWO algorithm is activated to commence the process. It consistently

updates the MPPT reference voltage signal to control method until the deviation of the optimal weed position is within 1%. The flowchart in Figure 3 illustrates the hybrid P&O-MIWOA approach.

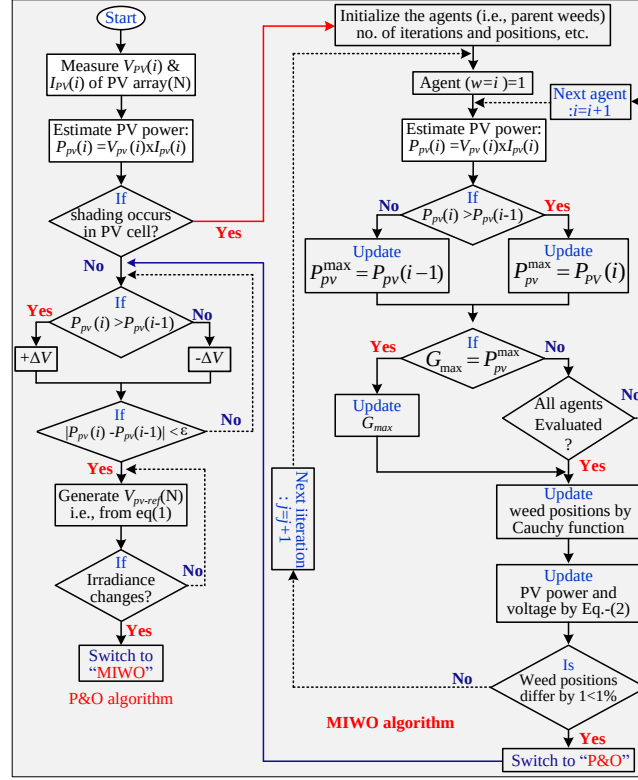


Figure 3: MIWOA-P&O Flow chart.

The Cauchy mathematical density function is used to express the MIWOA, which has two identical location/position and scale [19]. The MIWOA is capable of efficiently handling complex nonlinear functions in a multi-dimensional set. MIWOA is a powerful tool that excels at solving intricate optimization problems, including the challenging task of searching for complex functions. Its ability to converge rapidly ensures that it can identify the best solution efficiently. As a result, MIWOA is the perfect choice for identifying the GMP in PV systems operating under PSCs. During the search process, the newly born child weeds are strategically distributed throughout the search area, taking into account the position of the parent. The spread of these child weeds is determined by a standard deviation that varies according to equation 6 [19].

$$\sigma_{iter} = \frac{(iter_{max} - iter)^n}{iter_{max}^n} (\sigma_{initial} - \sigma_{final}) + \sigma_{final} \quad (6)$$

The variables $iter_{max}$ and $iter$ denote the upper limit and iterations for the generation's number of the function. The variables σ_{iter} : standard deviation, $\sigma_{initial}$: initial generation, and σ_{final} : final generation.

Upon cultivating a thriving tree, the community is allowed to produce novel weed. The fitness function may be estimated for newly emerged weed. It gradually diminishes from the highest permissible number of weeds (S_{max}) to the lowest acceptable number of weeds (S_{min}). This applies to both the most promising weed candidate and the least fit weed candidate in the population.

$$n(w_i) = \frac{S_{min}(fit(\omega_i) - \min fit) + S_{max}(\max fit - fit(\omega_i))}{\max fit - \min fit} \quad (7)$$

In this context, the constant parameter is the weed's produced seeds denoted by $n(w_i)$. The i^{th} weed fitness value is represented by $fit(w_i)$, while $\min fit$ signifies the fitness values corresponding to maximum and minimum populations. The P&O-MIWOA hybrid algorithm is capable of efficiently producing the reference voltage signal (V_{mpp}) under even PSC.

The control mechanism of proposed MPPT system is shown in Figure 4.

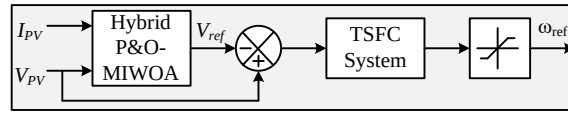


Figure 4: DC-link voltage controller with TSFC

The TSFC is implemented based on voltage error (x_i) and its derivative ($\dot{x}_i$) signal's membership functions as shown in Figure 5. The TSFC block fuzzifies the x_i and $\dot{x}_i$ input signals using two linguistic membership values, namely negative (N) and positive (P).

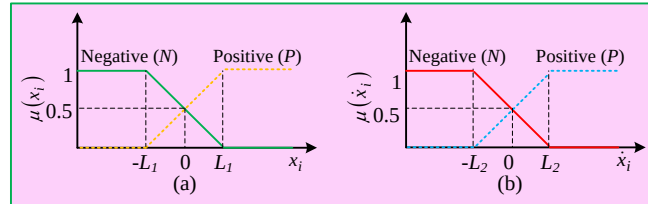


Figure 5: TSFC membership functions.

Table-3: The TSFC rule:

Rule	$x_i (m)$	$\dot{x}_i (m)$	Values
Rule-a	N	N	$Z_1 = d_1 x_i(m) + d_2 \dot{x}_i(m)$
Rule-b	N	P	$Z_1 = d_3 Z_1$
Rule-c	P	N	$Z_3 = d_4 Z_1$
Rule-d	P	P	$Z_4 = d_5 Z_1$

The TSFC corresponding rules can be found in Table-3. The parameters utilized in TSFC require tuning, while the rules Z1, Z2, Z3, and Z4 represent the consequent of the TSFC. The sampling instant is denoted by k. The fuzzy constants, d_1 , d_2 , d_3 , d_4 , and d_5 , are adjusted through a tuning process. The TSFC's output function (Y) is acquired through the utilization of a generalized defuzzifier, which is assessed using equation (8).

$$Y = \frac{Z_1 W_1 + Z_2 W_2 + Z_3 W_3 + Z_4 W_4}{Z_4 + Z_3 + Z_2 + Z_1} \quad (8)$$

$$\text{where, } W_1 = \min. \{ \mu_p(\dot{x}_i) \text{ and } \mu_p(x_i) \}, W_2 = \min. \{ \mu_p(x_i) \text{ and } \mu_N(\dot{x}_i) \}, \\ W_3 = \min. \{ \mu_N(x_i) \text{ and } \mu_p(\dot{x}_i) \}, W_4 = \min. \{ \mu_N(\dot{x}_i) \text{ and } \mu_N(x_i) \}$$

a. Modeling of BLDCM, MRAC and SMCS [4, 27-33]

The BLDC motor's rotor contains a permanent magnet, eliminating the need for a power supply to the rotor. Additionally, it exhibits a trapezoidal back EMF. Normally, only two stages of the BLDCM receive power simultaneously [28]. Provided are the basic formulas utilized in the modeling of the BLDCM. Presented below are the basic expressions used for modeling the BLDCM.

$$v_{ab} = R(i_a - i_b) + e_a - e_b + L \frac{d}{dt}(i_a - i_b) \quad (9)$$

$$v_{bc} = R(i_b - i_c) + e_b - e_c + L \frac{d}{dt}(i_b - i_c) \quad (10)$$

$$v_{ca} = R(i_c - i_a) + e_c - e_a + L \frac{d}{dt}(i_c - i_a) \quad (11)$$

$$T_e = \frac{i_a e_a + i_b e_b + i_c e_c}{\omega} \quad (12)$$

The line voltages in this system are referred to as v_{ab} , v_{bc} , and v_{ca} . The back EMFs of the BLDC motor's three phases are e_a , e_b , and e_c . The motor is supplied with three phase currents, which are i_a , i_b , and i_c . The winding

resistance is denoted by 'R', and the difference between the inductance of the stator winding (L_s) and the mutual inductance of the BLDCM (L_m) is represented by L . The speed of the motor is represented by

$$\text{Generally, } i_a = -i_b \quad \text{and} \quad i_c = 0 \quad \text{and also } e_c = 0 \quad \text{and} \quad e_a = -e_b \quad (13)$$

$$\text{Hence, } T_e = \frac{2i_a e_a}{\omega_r} \quad (14)$$

The back EMFs of a BLDCM are directly proportional to the speed of the BLDCM due to the presence of permanent magnets in the rotor. This relationship is commonly denoted by.

$$e_a = k_e \omega \quad (15)$$

$$\omega = \frac{v_{ab} - 2Ri_a - 2L \frac{di_a}{dt}}{2k_e} \quad (16)$$

$$\text{In steady state, } \frac{di_a}{dt} = 0 \quad \text{therefore, } \omega = \frac{v_{ab} - 2Ri_a}{2k_e} \quad (17)$$

The equations for the 'DQ' reference frame of a BLDCM will be represented as:

$$\Psi_{ds} = L_{ds}i_{ds} + L_{ad}i_{dr} + \Psi_{fd} \quad (18)$$

$$\Psi_{qs} = L_{qs}i_{qs} + L_{aq}i_{qr} \quad (19)$$

$$\Psi_{dr} = L_{dr}i_{dr} + L_{ad}i_{ds} + \Psi_{fd} \quad (20)$$

$$\Psi_{qr} = L_{qr}i_{qr} + L_{aq}i_{qs} \quad (21)$$

Where, $L_{dr} = L_{rl} + L_{ad}$, $L_{qr} = L_{rl} + L_{aq}$, $L_{ds} = L_{ad} + L_{sl}$, $L_{qs} = L_{sl} + L_{aq}$, here, L_{sl} and L_{rl} are leakage inductances of rotor and stator. L_{ad} and L_{aq} are armature inductances of the dq-axis. Ψ_{fd} is the permanent magnetic flux linkage.

The control of the speed of a BLDCM heavily relies on the i_{ds} and i_{qs} . These currents experience significant oscillations as the motor stabilizes at its desired speed. Therefore, the trajectory of the i_{ds} and i_{qs} is analyzed in the results section to assess their behavior under varying irradiance conditions. For detailed specifications of the BLDCM-pump motor, please refer to Table-4.

Table-4: Parameters of BLDCM [34-36]

S.No	Parameters	Values
1	Power.	5.0hp
2	Voltage.	380 – 440V _{rms}
3	Current.	7.5Amp
4	Speed.	1500rpm
5	DC voltage.	250-850V
6	Start up voltage.	500V _{dc}
7	Minimum and Maximum RPM.	600 to 2950
8	Maximum peak power.	5.0kW
9	Rated torque.	31Nm
10	Leakage reactance.	0.8Ohm
11	Inertia constant.	0.012kgm ²
12	Phase resistance.	0.9 Ohm
13	Magnetizing inductance.	10.5mH

14	Leakage inductance.	2.5mH
15	Magnetizing reactance.	3.3Ohm

The fundamental concept behind MRAC is to implement a closed loop control system that can modify the response of the modeled system by updating the motor/system parameters [29]. In order to estimate the speed of a BLDCM, the MRAC model utilizes the following equations.

$$e_a = \frac{k_e}{2} \omega_m F(\theta_e) \quad (22)$$

$$e_b = \frac{k_e}{2} \omega_m F(\theta_e - \frac{2\pi}{3}) \quad (23)$$

$$e_c = \frac{k_e}{2} \omega_m F(\theta_e + \frac{2\pi}{3}) \quad (24)$$

$$T_e = \frac{k_t}{2} \left[F(\theta_e) i_a + F(\theta_e - \frac{2\pi}{3}) i_b + F(\theta_e + \frac{2\pi}{3}) i_c \right] \quad (25)$$

$$\frac{d}{dt} \begin{pmatrix} i_a \\ i_b \\ \omega_m \\ \theta_m \end{pmatrix} = \begin{pmatrix} -R/L & 0 & 0 & 0 \\ 0 & -R/L & 0 & 0 \\ 0 & 0 & -K_f/j & 0 \\ 1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} i_a \\ i_b \\ \omega_m \\ \theta_m \end{pmatrix} + \begin{pmatrix} 2/3L & 1/3L & 0 \\ -1/3L & 1/3L & 0 \\ 0 & 0 & 1/J \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_{ab} - e_{ab} \\ v_{bc} - e_{bc} \\ T_e - T_{pm} \end{pmatrix} \quad (26)$$

$$\frac{d\omega_{ref}}{dt} = -a_m \omega_{ref} + b_m v_x \quad (27)$$

$$T_e = \sum_{n=1}^3 \frac{2}{3} I_m M i_m \sin^2(2\theta + \pi_n) = I_m \hat{M} i_n \quad (28)$$

$$\frac{de}{dt} = -a_m e + \left(a_m - \frac{B}{j}\right) \omega + \frac{\hat{M} i_m}{j} I_m^* - b_m v_x + \frac{T_{pm}}{j} + \frac{1}{j} \sum_{n=1}^{N_r} (g_k \cos(2k\theta) + h_n \sin(2n\theta)) \quad (29)$$

The constant of inertia is denoted as 'J' and the friction coefficient is represented as 'k_f'. The constants 'a_m, h_n, M, b_m, g_k, k', and 'n' are fixed parameters. Lastly,

$$I_m^* = \frac{J}{\hat{M} i_m} \left[b_m v_x + \left(\frac{B}{j} - a_m\right) \omega - \frac{T_{pm}}{j} - \frac{1}{j} \sum_{n=1}^{N_r} (g_k \cos(2k\theta) + h_n \sin(2n\theta)) \right] \quad (30)$$

The MRAC model of fundamental structure can be observed in Figure 6(a), while the BLDCM speed estimation is illustrated in Figure 6(b). The MRAC system's output is cross-referenced with the expected response derived from motor reference equations. The parameters utilized for speed estimation are adjusted using the PI by tracking the error input.

Typically, the voltage at dc-link is established based on the power generated by the PV. Likewise, the speed of the BLDCM relies on the input power for a specific torque. In this case, the torque due to load is remains constant under steady state conditions for a given depth of water. Consequently, the DCLC achieves the desired reference speed, as depicted in Figure 4. The estimated speed of the BLDCM is compared to the reference speed using the MRAC model, and the SMC generates the required reference current, as depicted in Figure 7. The implementation of SMCS relies on [31-32]. To acquire the necessary pulses for the converter, the actual and reference dc-link currents, along with the rotor position, are taken into account. This is demonstrated in Figure 8. The constants utilized in the SMC can be located in Table-5, whereas Table-6 provides the gains employed in the PI and the parameters of the TSFC system.

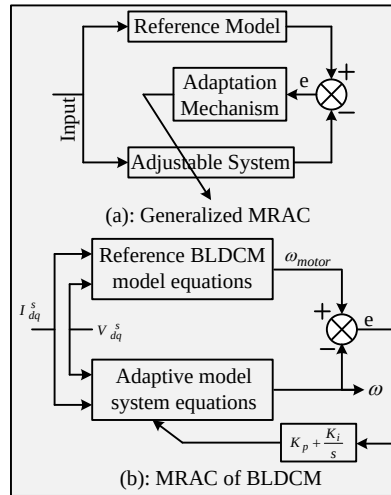


Figure 6: MRAC model configuration of BLDCM.

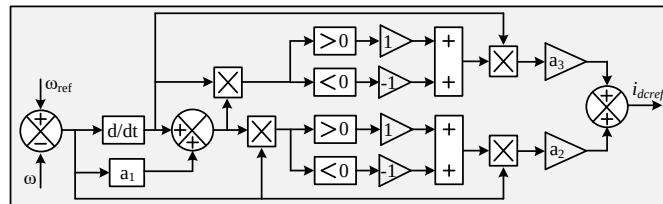


Figure 7: Block diagram of SMCS.

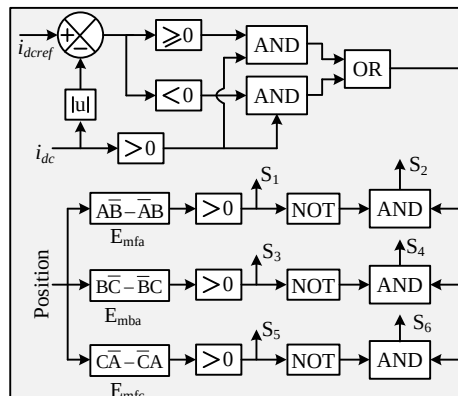


Figure 8: Switching states for converter.

Table-5: Constants of SMCS

S.No	Constants	Values
1	a_1	550
2	a_2	352
3	a_3	328

Table-6: Values of PI and TSFC

Controller	Value
TSFC	$a_1=3.9, a_2=3.25, a_3=-1.6, a_4=1.8, a_5=4.002$
PI gains of DCLC	$K_p=0.26, K_i=48.02$
PI gains of MRAC	$K_p=5.64, K_i=169.02$

III. HIL BASED RESULTS

The controllers that were suggested have been executed on real-time simulators (RTSs) devices, and the outcomes are summarized in this report on the OPAL-RT basis. Additional information about the RTS can be found in reference [39-40]. In order to establish the HIL configuration, two OPAL-RT units are utilized along with their respective computers. One unit functions as a WPS with inverter, PV, BLDCM, and pump as illustrated in Figure 1. Another unit is specifically designated for running the controllers associated with this model. Figure 9 illustrates the setup involving HIL configuration with OPAL RT modules. Analog signals are transferred from WPS (OPAL RT1) to the control module (OPAL RT2), whereas digital signals are transmitted from the control module to the pumping model. The OPAL-RT units communicate in real-time using both digital and analog signals, creating a HIL connection between OPAL-RT-1 and OPAL-RT-2 for efficient monitoring of real-time dynamics [39-40]. Various scenarios have been examined to showcase the efficiency and validate the functionality of the controllers that were created.

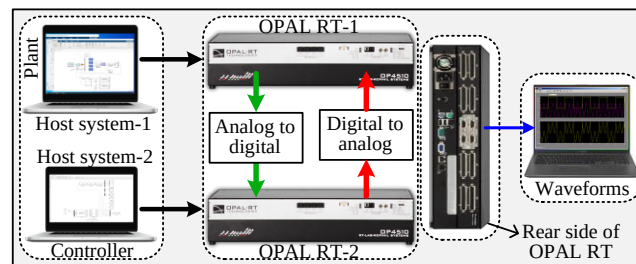


Fig. 9: HIL configuration model.

A. Case-A: Response of TSFC over PI

In this particular situation, the contemplation revolves around a shift in irradiance, transitioning from 1.0k to 750W/m^2 at $t = 3.0\text{sec}$. The performance of DCLC is being evaluated using both TSFC and PI, as illustrated in Fig. 5. While the gains of the PI controller are optimized for an irradiance of 1000W/m^2 , they do not perform as effectively when the irradiance drops to 750W/m^2 . On the other hand, the TSFC demonstrates the capability to adjust the output based on changes in irradiance. The actual and reference powers generated by the PV unit with both PI and TSFCs are presented in Figure 10. The results depicted in Fig. 10 show that the MPPT performance is superior with the TSFC compared to the PI controller in terms of extracting more power from the PV unit. This highlights the significant advantage of utilizing TSFC in DCLC. By effectively controlling the voltage at dc-link with the TSFC over the PI controllers, the PV system can generate slightly more power. Additionally, the oscillations in the voltage at dc-link due to the gains of the PI controller not being optimized for an irradiance of 750W/m^2 can impact the tracking of more energy from the PV system. Therefore, further investigations are conducted using the TSFC-based DCLC.

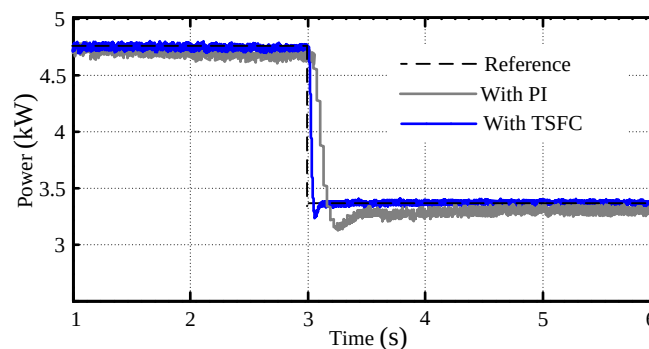


Figure 10: Performance of MPPT with PI and TSFC.

B. Case-B: Performance with various optimization methods

The primary issue lies in monitoring the maximum power output when the PV system is affected by PSC. A highly efficient proposed MIWOA has been created and compared against GWO, PSO and GA all of which are

combined with the P&O mechanism for the specified PSC in Table-2. The PV curves for these PSC can be seen in Figure 2, while the comparative power generated by the PV unit is shown in Figure 11. In this scenario, a consistent irradiance is assumed until $t=3.0\text{sec}$, PSC-1 occurs between $t=3.0\text{sec}$ and 5.0sec , PSC-2 starts at $t=5.0\text{sec}$ and ends at $t=7.0\text{sec}$, and PSC-3 takes place between $t=7$ to 9.0sec . As depicted in Figure 11, it is evident that the hybrid MIWOA outperforms GA, PSO, and GWO in terms of extracting maximum power from the PV unit while ensuring efficient tracking. Therefore, utilizing the proposed hybrid MIWOA can result in harvesting more energy from the PV unit. Figure 12 depicts the voltage at dc-link associated with PSCs, shown next to MIWOA.

The bar graph depicted in Figure 13 showcases the comparative tracking responses of the maximum power output of the PV unit under PSCs. It is evident from the figure that the hybrid MIWOA proposed shows a notable advantage and performs effectively during PSC compared to GWO, PSO and GA. This is due to its capability to swiftly track V_{mpp} in comparison to other optimization techniques.

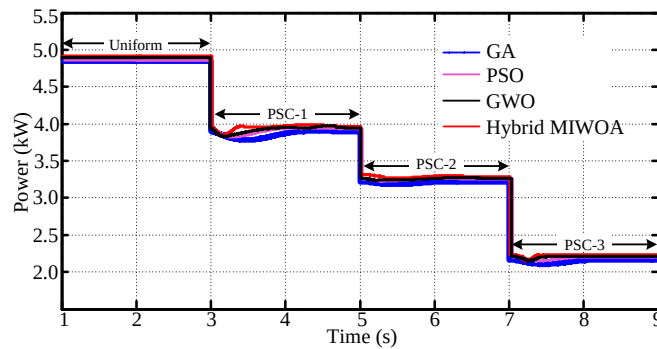


Figure 11: Powers from various methods under PSCs.

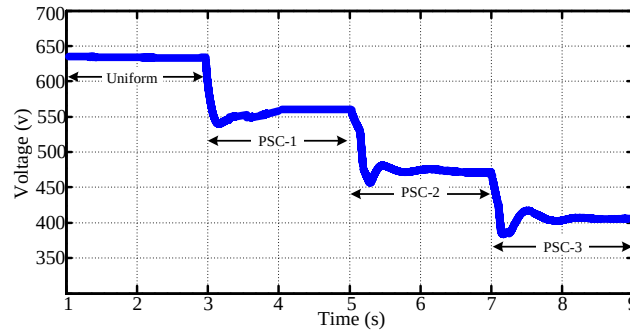


Figure 12: Voltage at dc-link with MIOWA.

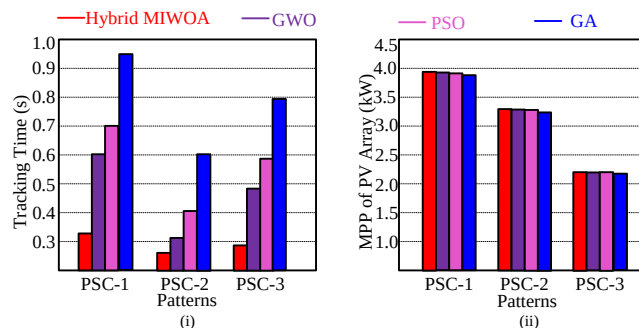


Figure 13: bar chat with MIWOA, GWO, PSO and GA.

C. Case-C: Torque, Speed and discharge of water during change in solar irradiances

The WPS undergoes testing when the irradiance changes suddenly from maximum (i.e., 1 kW/m^2) to minimum (i.e., 0.4 kW/m^2) at $t=3.0\text{ sec}$. As the irradiance decreases abruptly, the power availability from the PV also decreases, leading to the DCLC generating the corresponding reference speed for the BLDCM. The MRAC is capable of estimating the speed of the BLDCM. The inverter receives pulses from the SMCS by comparing these two speed signals. SMCS outperforms the PI based speed controller, resulting in a more precise tracking response of the motor speed. The speed response is depicted in Figure 14. The prioritized approach for attaining the desired speed promptly is demonstrated in Figure 14, utilizing the proposed SMCS-based method. The presence of additional ripples in the motor speed is observed when using the PI based controller compared to SMCS. 1. Nevertheless, the motor's torque stays consistent, while the electromagnetic torque produced by the BLDCM is illustrated in Figure 15. Additionally, as the motor speed decreases, the flow rate of water through the pump decreases, resulting in the corresponding water discharge shown in Figure 16.

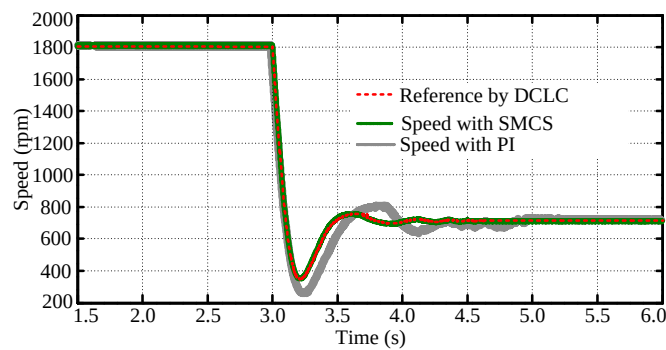


Figure 14: Speed of BLDCM with SMCS over PI.

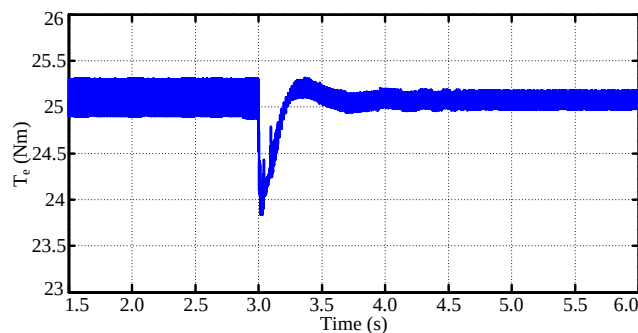


Figure 15: Torque of BLDCM.

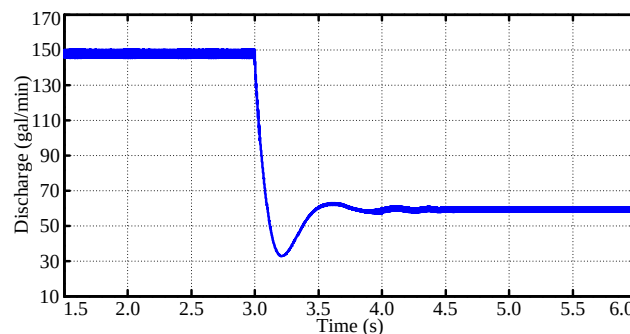


Figure 16: Water discharge.

IV. CONCLUSION

A hybrid MIWOA has been utilized to perform PSC on a PV-based WPS that is powered by a speed sensor-less BLDCM and supported by SMCS. The effectiveness of this approach is compared to GWO, PSO, and GA in monitoring the MPPT performance during PSC of the PV unit. Various practical scenarios are demonstrated on

the OPAL-RT platform. The integration of TSFC based DCLC enables a rapid tracking response of the voltage at dc-link. The use of TSFC clearly shows a significant capability in adjusting the output according to MPPT in contrast to the PI. The suggested model is highly recommended for WPS, even under PSCs, and addresses practical challenges such as the inclusion of an additional pipeline for water transfer between locations, while also considering the practical load torque characteristics on the BLDCM. Comprehensive results are provided to validate the proposed approach on the OPAL-RT platform. The hybrid MIWOA has been noted to enable the extraction of a greater amount of energy from the PV unit. The proposed model is designed without the need for energy storage systems like batteries and additional DC-DC converters, making it a cost-effective MPPT solution for the PV system. The inverter functions as the MPPT device of the PV unit with the proposed control of the inverter, and an MRAC controller is employed to eliminate the requirement for sensors to detect the BLDC motor's speed.

ACKNOWLEDGMENT

I would like to express my sincere gratitude to Dr. Siva Ganesh Malla, Director ERG foundation for his valuable guidance, encouragement and support throughout the research process. His expertise and mentorship have been instrumental in shaping the direction and quality of this work.

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